

ChE 311 – Transport Phenomena II –Spring 2024

Rubric Assessment Summary

The assessment of ABET Student Outcome 1 was conducted using the final exam as the primary direct assessment instrument. The exam was scored out of 100 points.

ABET Outcome 1

An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Performance Indicator (PI)	Level 3: Satisfactory <30	Level 2: Developing 30 to 60	Level 1: Unsatisfactory >60
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems
6. The uses of appropriate resources needed to solve problems	Uses appropriate resources to locate information needed to solve problems	Uses limited resources to solve problems	Uses no resources to solve problems

FINAL EXAM TOTAL POINTS 100

Average Score	41.67
Standard dev.	27.17

RESULTS BY PERFORMANCE INDICATOR

Rubric #	1
Mapped SO	1
Average	1.88
Std Dev	0.85
S	7
D	7
U	10

Summary of Findings

The final exam (total points = 100) was used as a direct assessment tool for **Student Outcome 1 (SO1)**. The overall class performance was low, with an **average score of 41.67** and a **standard deviation of 27.17**, indicating both weak achievement and high variability among students.

For **Performance Indicator (Rubric #1)** mapped to SO1, the **average score was 1.88** (on the rubric scale), with a **standard deviation of 0.85**. The distribution of performance levels shows that **7 students met expectations (Satisfactory)**, **7 were Developing**, and **10 were Unsatisfactory**. This means that a majority of students (17 out of 24) did not reach satisfactory attainment.

Interpretation

These results indicate that **SO1 is not being adequately attained**. The low mean score and high proportion of unsatisfactory ratings suggest that students are struggling significantly with the fundamental skills and knowledge associated with this outcome.

Recommended Improvements

To improve attainment of SO1, the following actions are recommended:

- Review alignment between course content, instruction, and assessed material.
- Incorporate additional formative assessments to monitor student understanding earlier in the term.
- Provide targeted support (e.g., tutoring sessions, review workshops) for at-risk students.
- Reevaluate exam design and rubric clarity to ensure consistency with expected performance levels.

- Consider pedagogical adjustments such as active learning strategies or scaffolded problem-solving exercises.

Conclusion

While **Student Outcome 1 is not achieved, the assessment indicates poor performance, with significant variability among students.**

The above actions will be reassessed in the next offering of the course to determine their effectiveness in improving student performance and outcome attainment.

TRANSPORT PHENOMENA II

CHE 312, SPRING 2024

FINAL EXAM

CLOSED BOOK, TIME 1 HR 45 MIN

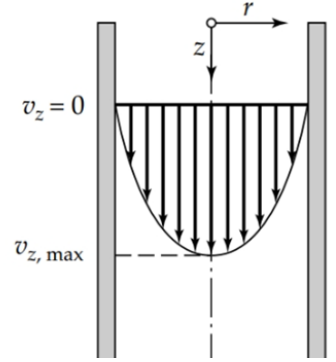
- 1- A highly viscous oil is in laminar flow in a circular tube of inner radius R according to the following velocity profile:

$$v_z = v_{z,max} \left[1 - \left(\frac{r}{R} \right)^2 \right]$$

The inside wall of the tube is held at a constant temperature T_0 . For a “thermally fully developed” flow (after fluid is sufficiently far downstream), the thermal energy generated by viscous dissipation is being entirely conducted out through the pipe wall, and thus, the temperature profile of the liquid is independent of z .

Answer the following questions in terms of the following constants:

$v_{z,max}$, the inner radius of the tube R , the viscosity of the fluid μ , the thermal conductivity of the fluid k , and the inner wall temperature T_0 .



- a. Use the dissipation function Φ_v to calculate the rate of viscous dissipation of energy per unit length of the tube. (10 points)

$$\Phi_v = \left(\frac{dv_z}{dr} \right)^2 = \frac{4v_{z,max}^2}{R^4} r^2$$

$$\int_0^R dr \int_0^{2\pi} r d\theta \frac{4\mu v_{z,max}^2}{R^4} r^2 = 2\pi \frac{4\mu v_{z,max}^2}{R^4} \int_0^R dr r^3 = 2\pi \mu v_{z,max}^2$$

- b. Using the proper energy balance equation, derive an ordinary differential equation for the fully developed temperature profile of the fluid as a function of radius. (10 points)

$$0 = k \left[\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) \right] + \mu \frac{4v_{z,max}^2}{R^4} r^2$$

- c. Integrate the above differential equation using the proper boundary conditions and derive the thermally fully developed temperature profile of the fluid as a function of radius. (10 points)

$$\begin{aligned} \frac{d}{dr} \left(r \frac{dT}{dr} \right) &= -\frac{4\mu v_{z,max}^2}{kR^4} r^3 \\ r \frac{dT}{dr} &= -\frac{\mu v_{z,max}^2}{kR^4} r^4 + C_1 \rightarrow \frac{dT}{dr} = -\frac{\mu v_{z,max}^2}{kR^4} r^3 + \frac{C_1}{r} \end{aligned}$$

$\frac{dT}{dr}$ is finite at $r = 0$, thus $C_1 = 0$.

$$T = -\frac{\mu v_{z,max}^2}{4kR^4} r^4 + C_2$$

$$T(r = R) = T_0 \rightarrow C_2 = T_0 + \frac{\mu v_{z,max}^2}{4k}$$

$$T = T_0 + \frac{\mu v_{z,max}^2}{4k} \left[1 - \left(\frac{r}{R} \right)^4 \right]$$

- d. Using the above temperature profile, calculate the radial heat flux on the inner surface of the pipe. (10 points)

$$q_r|_{r=R} = -k \frac{dT}{dr} \Big|_{r=R} = \frac{\mu v_{z,max}^2}{R}$$

- e. Calculate the bulk temperature of the thermally fully developed fluid. (10 points)

$$\begin{aligned} T_b &= \frac{\int_0^R dr \int_0^{2\pi} r d\theta v_z T}{\int_0^R dr \int_0^{2\pi} r d\theta v_z} \\ \int_0^R dr \int_0^{2\pi} r d\theta v_z &= \frac{\pi R^2}{2} v_{z,max} \\ \int_0^R dr \int_0^{2\pi} r d\theta v_z T &= 2\pi \int_0^R dr r v_{z,max} \left[1 - \left(\frac{r}{R} \right)^2 \right] \left(T_0 + \frac{\mu v_{z,max}^2}{4k} \left[1 - \left(\frac{r}{R} \right)^4 \right] \right) \\ &= \frac{\pi R^2}{2} v_{z,max} T_0 + 2\pi \frac{\mu v_{z,max}^3}{4k} \int_0^R dr r \left[1 - \left(\frac{r}{R} \right)^2 \right] \left[1 - \left(\frac{r}{R} \right)^4 \right] \\ &= \frac{\pi R^2}{2} v_{z,max} T_0 + 2\pi \frac{\mu v_{z,max}^3}{4k} \frac{13}{56} R^2 \\ T_b &= T_0 + \frac{5}{24} \frac{\mu v_{z,max}^2}{k} \end{aligned}$$

- f. Calculate the Nusselt number defined as $Nu = \frac{2hR}{k}$, for the thermally fully developed fluid.

(10 points)

$$\begin{aligned} q_r|_{r=R} &= h(T_b - T_0) \rightarrow h = \frac{\frac{\mu v_{z,max}^2}{R}}{T_b - T_0} = \frac{\frac{\mu v_{z,max}^2}{R}}{\frac{5}{24} \frac{\mu v_{z,max}^2}{k}} = \frac{24}{5} \frac{k}{R} \\ Nu &= \frac{48}{5} = 9.6 \end{aligned}$$

- 2- Solid species A in the shape of a sphere of radius R is submerged in a large body of stagnant liquid B . At the given temperature, species A is slightly soluble in the liquid B with equilibrium concentration of c_{A0} . The dissolved species A diffuses in the liquid phase with constant diffusivity D_{AB} and it undergoes a first order reaction with rate constant k_1''' .

Answer the following questions in terms of the following constants: radius of the solid sphere R , the diffusivity D_{AB} , and the rate constant k_1''' .

- a- By means of a shell mole balance or the appropriate equation of continuity for species A , derive an ordinary differential equation for the quasi-steady-state concentration profile of species A in the liquid phase. Hint: $x_A \ll 1$ (20 points)

$$D_{AB} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dc_A}{dr} \right) - k_1''' c_A = 0$$

- b- Solve the above differential equation and express the quasi-steady-state radial concentration profile of species A in terms of the Thiele modulus $\phi \equiv \sqrt{\frac{k_1'''}{D_{AB}}} R$. (10 points)

Hint: $\frac{1}{r} \frac{d}{dr} \left(r^2 \frac{dc_A}{dr} \right) = \frac{d^2}{dr^2} (rc_A)$

$$\begin{aligned} \frac{d^2}{dr^2} (rc_A) &= \frac{k_1'''}{D_{AB}} rc_A \\ c_A &= \frac{C_1}{r} \exp \left(\sqrt{\frac{k_1'''}{D_{AB}}} r \right) + \frac{C_2}{r} \exp \left(-\sqrt{\frac{k_1'''}{D_{AB}}} r \right) \\ c_A &= \frac{C_1}{r} \exp \left(\phi \frac{r}{R} \right) + \frac{C_2}{r} \exp \left(-\phi \frac{r}{R} \right) \end{aligned}$$

Alternatively, the solution may be expressed in terms of hyperbolic functions:

$$c_A = \frac{C_1}{r} \sinh \left(\phi \frac{r}{R} \right) + \frac{C_2}{r} \cosh \left(-\phi \frac{r}{R} \right)$$

- c- Find the integration constants using the proper boundary conditions. (10 points)

Boundary condition 1: $c_A(r = \infty) = 0 \rightarrow C_1 = 0$

Boundary condition 2: $c_A(r = R) = c_{A0} \rightarrow C_2 = \frac{c_{A0} R}{e^{-\phi}}$

$$c_A = \frac{c_{A0} R}{e^{-\phi}} \frac{1}{r} \exp \left(-\phi \frac{r}{R} \right)$$

Alternatively,

$$c_A = \frac{c_{A0} R}{e^{-\phi}} \frac{1}{r} \left[\sinh \left(\phi \frac{r}{R} \right) + \cosh \left(-\phi \frac{r}{R} \right) \right]$$